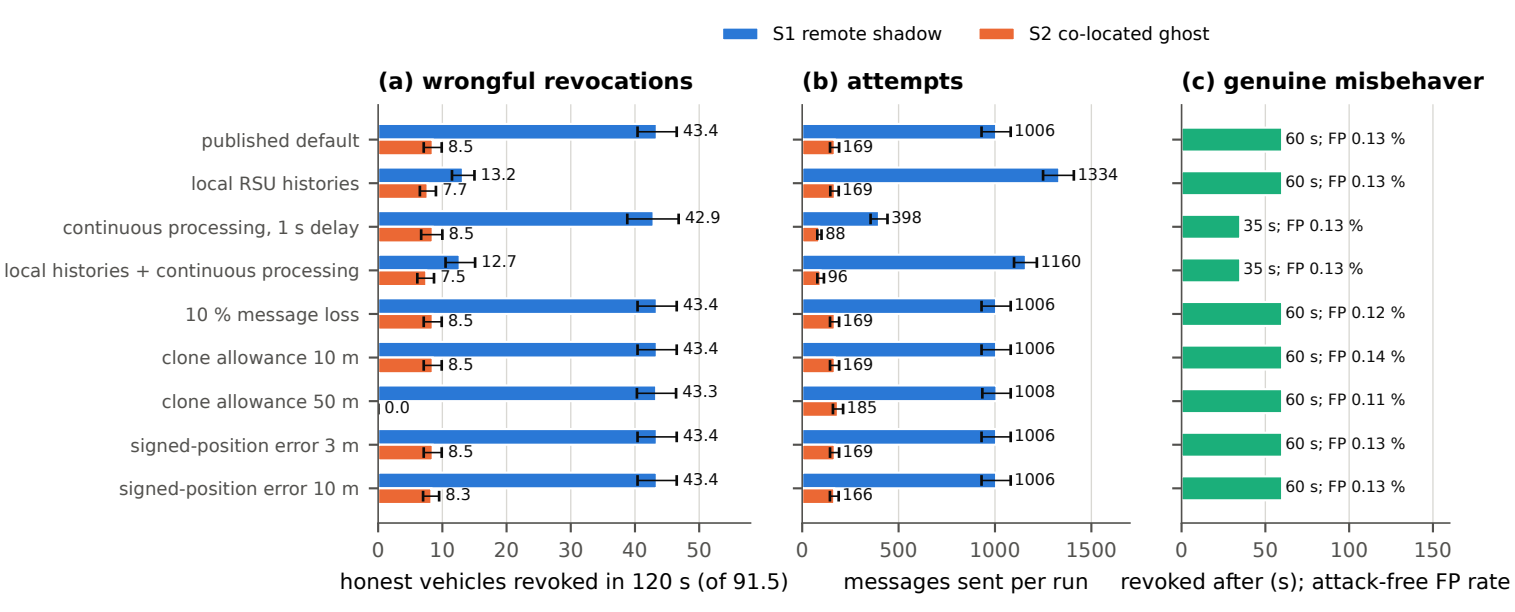
S1 remote shadow
S2 co-located ghost
(a) wrongful revocations
(b) attempts
(c) genuine misbehaver
published default
local RSU histories
continuous processing, 1 s delay
local histories + continuous processing
10 % message loss
clone allowance 10 m
clone allowance 50 m
signed-position error 3 m
signed-position error 10 m
43.4
8.5
13.2
7.7
42.9
8.5
12.7
7.5
43.4
8.5
43.4
8.5
43.3
0.0
43.4
8.5
43.4
8.3
1006
169
1334
169
398
88
1160
96
1006
169
1006
169
1008
185
1006
169
1006
166
60 s; FP 0.13 %
60 s; FP 0.13 %
35 s; FP 0.13 %
35 s; FP 0.13 %
60 s; FP 0.12 %
60 s; FP 0.14 %
60 s; FP 0.11 %
60 s; FP 0.13 %
60 s; FP 0.13 %
0
10
20
30
40
50
0
500
1000
1500
0
50
100
150
honest vehicles revoked in 120 s (of 91.5)
messages sent per run
revoked after (s); attack-free FP rate